

GAGAN VEGINATI

Machine Learning Engineer

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PROFESSIONAL SUMMARY

Machine Learning Engineer (M.S., Boston University) focused on taking ML models from prototype to deployed, reliable systems. Built graph neural networks, CNNs, and LLM-powered applications, and shipped a live AI oracle on AWS. Runner-up at the EasyA Consensus hackathon (Toronto, 2025). Strong in PyTorch, LLMs (LangChain, RAG), AWS, and MLOps.

TECHNICAL SKILLS

ML & Deep Learning: PyTorch, PyTorch Geometric, TensorFlow, Keras, Scikit-learn, XGBoost, CNNs, GNNs (GCN/GraphSAGE/GAT), LSTM, Computer Vision (OpenCV), NLP

LLMs & GenAI: Claude, Hugging Face Transformers, LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Vector Databases (FAISS/Chroma), Prompt Engineering

MLOps & Cloud: AWS (Lambda, Bedrock, SageMaker, EC2, S3), Docker, Kubernetes, MLflow, Weights & Biases, CI/CD, FastAPI, Redis, Apache Spark

Languages & Data: Python, SQL, Java, C/C++, R, Pandas, NumPy, Databricks, Power BI, Git, Linux

EXPERIENCE

Graduate Assistant

Jan 2025 – May 2025

Boston University — Department of Computer Science

Boston, MA

- Automated data processing workflows for 500+ students, cutting manual handling time by 30%
- Mentored students and presented analysis of large-scale academic datasets to faculty, supporting reporting for 2,000+ students

Data Research Intern

May 2023 – May 2024

SRM University AP — International Relations Department

Amaravati, India

- Developed ML models (Random Forest, XGBoost, Logistic Regression) predicting student retention with 89% accuracy across 15,000+ records
- Built an end-to-end pipeline processing 5GB+ of data with Apache Spark and Python, reducing ETL time by 65%

PROJECTS

Gavel — AI Oracle for Prediction Markets

May 2026

Claude (LLM), FastAPI, AWS Lambda/Bedrock, Solidity, Base, Next.js

- Built and deployed a live pay-per-query AI oracle that resolves prediction-market questions in ~14 seconds by reasoning over cited sources and returning a verdict with on-chain proof
- Engineered a serverless FastAPI backend on AWS Lambda with an AWS Bedrock failover path, plus a custom HTTP-402 (x402) micropayment layer

GNN-Based Stock Relations & Prediction

Sep – Dec 2025

Python, PyTorch Geometric, Streamlit

- Built correlation graphs of 29 large-cap U.S. stocks and trained GCN, GraphSAGE, GAT, and Temporal-GAT models against a Random Forest baseline
- Best model (GraphSAGE) reached 58.7% next-day directional accuracy and F1 = 0.654; shipped a Streamlit app with an attention-based peer-influence graph

Financial Transaction Anomaly Detection

Jan – May 2024

Python, Scikit-learn, SMOTE, KNN, Random Forest, LSTM

- Balanced highly imbalanced fraud data with SMOTE, improving F1 by 95% (relative) across KNN, Random Forest, and LSTM models
- Compared classical and deep-learning detectors and tuned thresholds to balance false positives against missed fraud

EDUCATION

Boston University — M.S. in Computer Science

Sep 2024 – May 2026

Coursework: Machine Learning, Data Science with Python, Advanced Programming Techniques, Algorithms

Boston, MA

SRM University AP — B.Tech in Computer Science (AI/ML Specialization)

Sep 2020 – May 2024

Coursework: Deep Learning, Computer Vision, Data Structures & Algorithms, Database Management

Amaravati, India

CERTIFICATIONS & LEADERSHIP

- Google Cloud – Prompt Design in Vertex AI; MathWorks – Image Processing & MATLAB Onramp
- Sports Secretary, SRM University AP — led a team of 30 and organized 18+ events for a 15,000-student campus